

ARYAN JAIN

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Education

National University of Singapore

Aug 2025 – Dec 2026

Master of Computing (Computer Science Specialisation)

Masters Dissertation — CVML Group, advised by Prof. Angela Yao

Vellore Institute of Technology, Vellore Campus

June 2021 – June 2025

Bachelor of Technology in Computer Science and Engineering

Amazon ML Summer School — Selected from thousands of applicants (July 2024)

Publications

HMS: Bounded Harmonic Mean Perturbations for Stabilizing SGD-Based Regression Optimization

Submitted to AAAI 2027

- Helped implement the HMS optimization module and integrated it into multiple optimizers including SGD, Adam, and RMSprop.
- Built large-scale regression experiment pipelines and contributed to empirical evaluation across hundreds of training runs.
- Co-authored the manuscript as the fourth author in collaboration with researchers from Stanford University and University of Maryland.

Experience

National University of Singapore

Jan 2026 – Present

Researcher, CVML Group, advised by Prof. Angela Yao

- Developing agentic multimodal LLM systems for streaming video understanding and reasoning.

Siemens

April 2026 – Aug 2026

AI Research Intern, Foundational Technologies Research and Pre-Development

- Fine-tuned **TinyLLaVA-1.5B** on TextVQA and built an epoch-level probing pipeline tracking attention and embedding geometry across 22 layers on a fixed set of 5 benchmark images.
- Extended the pipeline to **Qwen3.5-4B** with custom hooks isolating per-head attention (16 heads $\times$ 8 layers) and ran per-head **Procrustes stability analysis** across 22 queries to quantify cross-layer representation drift.
- Designed a causal ablation suite (head zeroing, patch occlusion, attention-map transplantation) over 350+ regenerated responses, finding that attention magnitude, representational stability, and ablation sensitivity rank heads almost independently; exposing a gap in single-metric interpretability methods.

PayU

Dec 2024 – June 2025

Data Science Intern, Innovations and Finance Team

Bangalore, India

- Built an end-to-end Affluence Scorecard Model using regression, reducing customer acquisition costs by **20–25%**.
- Created a robust feature engineering pipeline on financial bureau data (**20M+ rows**), generating **2400+ features**. The resulting feature bed was foundational for multiple downstream models.
- Developed XGBoost models for early risk detection and default prediction, reducing collection costs by **40%** and achieving an AUC of **0.85**.
- Helped create **1,200+** behavioral features and conducted EDA to inform model strategy for credit loss prevention; performed QA to ensure correctness and reliability of model inputs.

IIT Guwahati

Sept 2023 – April 2024

Machine Learning Research Intern under Prof. Prabir Barooah

Guwahati, India

- Converted raw Excel data to a Pandas DataFrame, applying extensive preprocessing for accurate analysis.
- Implemented ML models to reconstruct missing power consumption data of IIT Guwahati, controlling for **170** parameters based on time and temperature.
- Achieved accuracy scores ranging from **0.6 to 0.8** across approaches including non-linear optimization, SVM, and linear regression.

Projects

Anchor – On-Device AI Wellness Companion

Dec 2025 – Present

- Building a privacy-first on-device AI wellness companion for mobile phones, fine-tuning **LLaMA 3.2 3B** with a QLoRA SFT pipeline and a **4-tier context injection system**; using the memory system on base model alone led to a **10%** boost in benchmark performance.
- Designed an **83-scenario eval framework** across 11 behavioral categories; best model achieved **61.5%** overall, beating the base model (**57.4%**).

SIAANet – Multi-Edge Graph Transformer for Stock Forecasting

Sep 2025 – Nov 2025

- Constructed a heterogeneous multi-edge graph over S&P 500 tickers using correlation matrices, PCMCI-derived causal graphs, industry-sector priors, and embedding-based similarity graphs.
- Engineered a custom Graph Transformer (**2.29M params**) with 1D convolutional temporal encoders, learnable node/edge-type embeddings, multi-edge attention, and **7 decoupled horizon-specific heads**.
- Achieved directional accuracy of **0.7764 (1-day)**, **0.7226 (2-day)**, **0.7121 (3-day)**, surpassing Chronos-T5-Small despite being **20× smaller**; average MAPE of **6.2%**, RMSE of **1.85**, MAE of **1.12**.

Technical Skills

Languages: Python, Java, C/C++, HTML/CSS, SQL/PLSQL

Tools/Libraries: PostgreSQL, FastAPI, TensorFlow, Numpy, Pandas, Figma, scikit-learn, Git, CasADi, PyTorch, OpenCV, AWS, LLM, LangChain

Certifications:

- **Computational Thinking for Problem Solving**
Issued by: University of Pennsylvania
- **Coding for Everyone: C and C++ Specialisation (set of 4 courses)**
Issued by: University of California, Santa Cruz
- **Supervised Machine Learning: Regression and Classification**
Issued by: Stanford and DeepLearning.AI
- **Advanced Learning Algorithms**
Issued by: Stanford and DeepLearning.AI
- **AWS Academy Graduate – AWS Academy Cloud Foundations Certificate**
- **AWS Academy Graduate – AWS Academy Cloud Architecting Badge**

Extracurricular

Chess

- Hold an international FIDE rating of **1629**.
- Represented NUS at international chess tournaments.
- Represented VIT at the Kasparov Chess Foundation Chess Tournament (international).
- Secured victory in the Chess Masters Premier League 2023 organized by IIT Kanpur.